

[illegible]

No.	Task /Activity	Hazard and Consequence		Initial Risk			Mitigating Controls	Residual Risk		
	Sequential Steps	Hazard	Consequence	Probability	Consequence Severity	Risk Rating	Measures to Reduce Risk to ALARP	Probability	Consequence Severity	Risk Rating
1	Review TRA and TBT	Miscommunication	Personnel Injury / Environmental Pollution	1	2	2	Situational awareness and proper use of PPE. Hard hats, impact gloves, cotton gloves, latex gloves, steel toed boots, safety glasses, side shields and face shield	1	1	1
2	Prior to any task being undertaken ensure that all personnel are aware of the hazards involved.	Electrical shock, possible slips trips and falls. Pinch points poor body position. Hydraulic discharge and spills.	Personnel Injury / Environmental Pollution	2	3	6	All personnel must be aware of their surrounds and weather conditions. Use proper Manual Handling techniques, i.e., Always use proper body positioning, lift with legs, and obtain second personnel when loads are over 40lbs. Read all SDS on chemicals used, use PPE that is specified in SDS and be aware of where all spill kits are located.	2	2	4
3	Obtain the proper testing equipment and tools needed for the task. Proper PPE	Damage to equipment, electrical shock.	Personnel Injury / Equipment Damage	3	3	9	Proper PPE Have a trained tech use test equipment. Use the proper tool for the task.	2	2	4
4	Electrical troubleshooting or repair	Electrical shock, damage to equipment	Personnel Injury / Equipment Damage	3	3	9	Ensure the system is grounded. Proper troubleshooting techniques are utilized.	2	2	4
5	<i>If Required a Lock out tag out electrical source preparing for testing. Review LOTO TRA. Obtain a PTW if needed from bridge.</i>	<i>Electrical shock, damage to equipment.</i>	Personnel Injury / Equipment Damage	2	2	4	<i>Review LOTO TRA. All personnel conducting work will always have keys for locks on them.</i>	1	1	1
6	Electrical troubleshooting or repair.	Electrical shock, damage to equipment	Personnel Injury / Equipment Damage	3	3	9	Ensure that NO WORK is completed on live circuits only for voltage testing.	2	2	4
7	Electrical troubleshooting or repair	Electrical shock, damage to equipment	Personnel Injury / Equipment Damage	3	3	9	Ensure the equipment / device is off before disconnecting cables.	2	2	4

8	Electrical troubleshooting or repair	Electrical shock, damage to equipment	Personnel Injury / Equipment Damage	3	3	9	Always work as a team, there should always be two personnel involved in all electrical troubleshooting and repairs.	2	2	4
9	Task completed closing all electrical connections and/or boxes.	Electrical shock, damage to equipment	Personnel Injury / Equipment Damage	2	2	4	Personnel are to close all electrical connects that they have been working on for that task.	1	1	1
10	Remove lock out tag out, ready for voltage to be re applied.	Electrical shock, damage to equipment.	Personnel Injury / Equipment Damage	2	2	4	Personnel will remove their locks from the lock out tag out. Before turning on circuit they must be sure that all personnel are clear, and circuit is hooked back up. Good communication between personnel. Close out PTW from vessel if one was obtained.	1	1	1
11	Stowing all tools and testing equipment away in its proper location after task is completed.	Damage to equipment, slips, trips and falls.	Personnel Injury / Equipment Damage	2	2	4	Stow all equipment in their proper location or cases and tools in their proper location. Clean area of all slips, trips, and falls. Put away and secure all ladders if used.	1	1	1

TRA Approval	Name	Position	Signature	Date
By signing, the Approver confirms that the TRA has been reviewed and considers the task acceptable to proceed with residual risk meeting the acceptance criteria of DO Risk Matrix.				

*Reference DO-HSE-FRM-140 for DeepOcean Risk Assessment Matrix (RAM) and Guidewords